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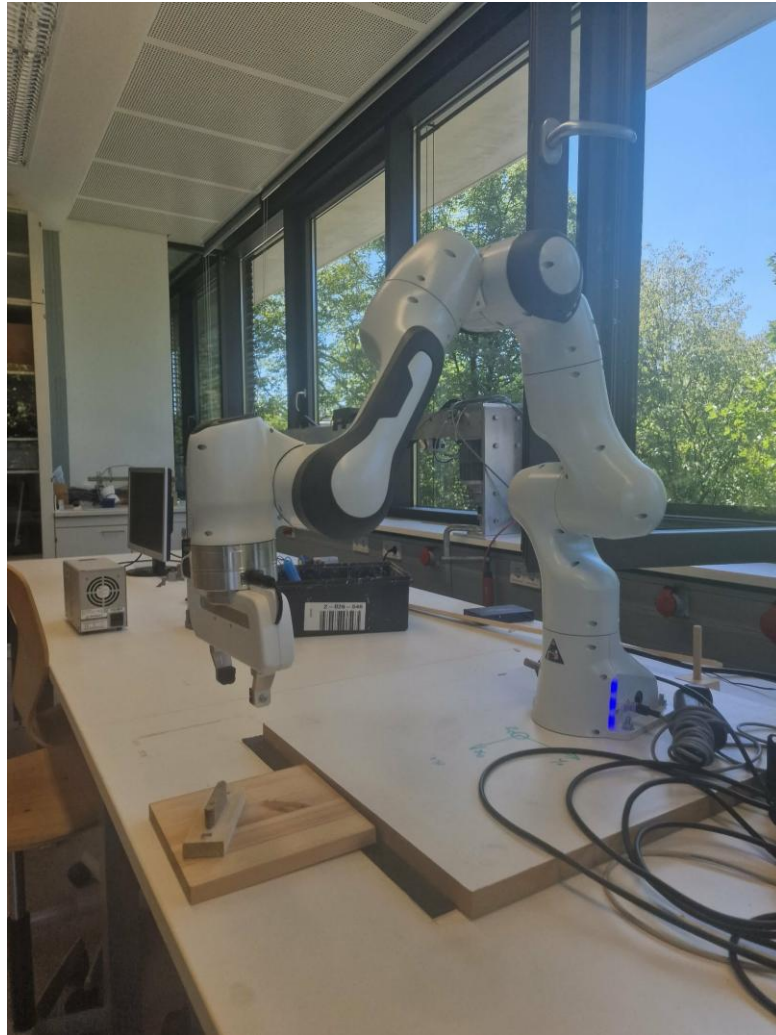
Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

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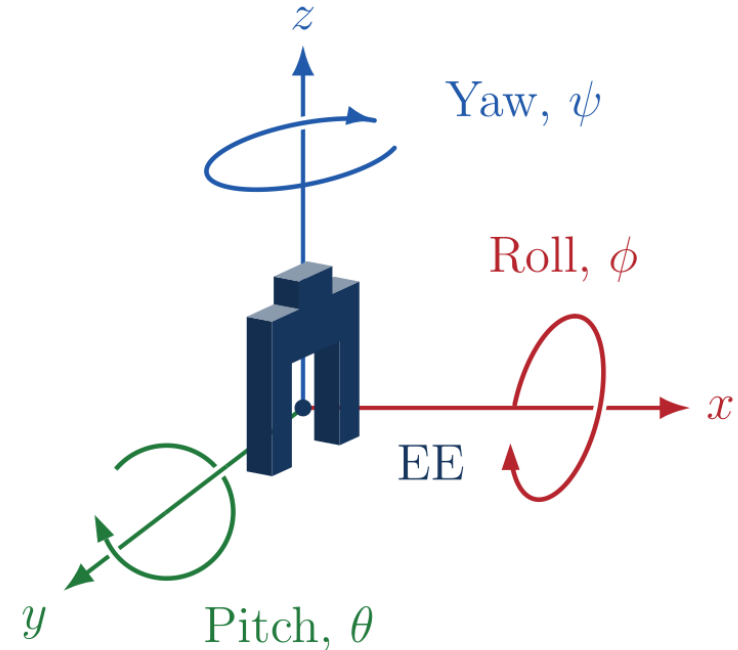
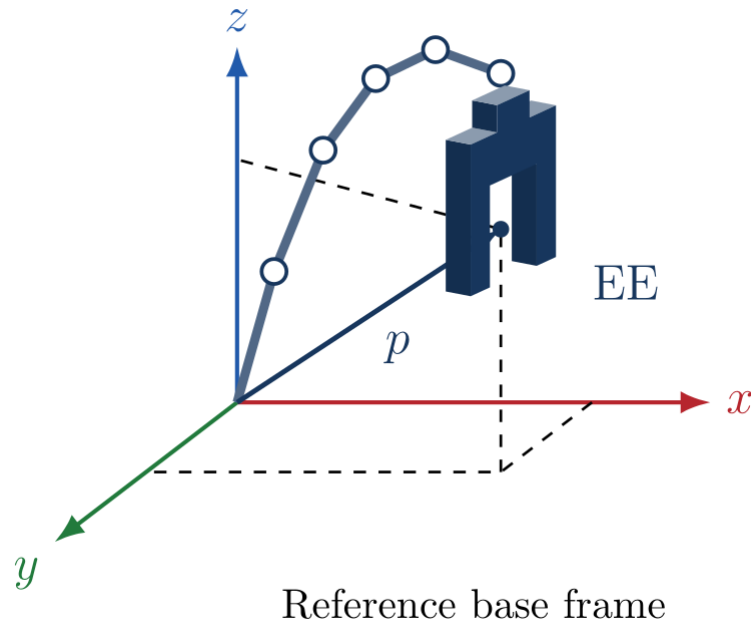
- Problem
 - Grinding requires an aligned tool face.
 - Calibration leaves uncertainty in the physical surface orientation.
- Idea
 - Cartesian impedance makes position and rotation compliant.
 - Pressing a tilted tool creates a contact moment.
 - Rotational compliance allows the tool to turn towards alignment.



1. Introduction
2. Cartesian controller
3. Contact experiments
4. Contact results
5. Null-space control and experiments
6. Conclusion and future work

Cartesian pose

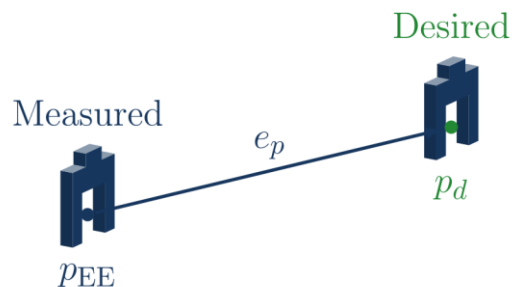
- The end effector (EE) is the tool or gripper at the end of the robot arm.
- Cartesian pose describes its position and rotation in 3D (6 degrees of freedom).
 - Position: 3 translations
 - Orientation: 3 rotations



- The EE position depends on the configuration of all 7 joint angles: $p = p(q)$

- Positional error

$$e_p = p_d - p_{EE}$$



$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

- Cartesian wrench

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & \mathbf{0} \\ \mathbf{0} & K_R \end{bmatrix}}_{K_C} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & \mathbf{0} \\ \mathbf{0} & D_R \end{bmatrix}}_{D_C} \begin{bmatrix} \dot{p}_d - \dot{p}_{EE} \\ \omega_d - \omega_{EE} \end{bmatrix}$$

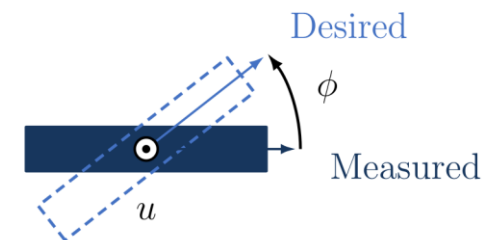
- In the surface frame, K and D are diagonal and decouple translation and rotation.

K_p, K_R translational / rotational stiffness

$\dot{p}, \omega$ linear / angular velocity

- Rotational error

$$e_R = \phi u$$



$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

- Joint-torque commands from Cartesian wrench

$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \quad \tau_{\text{cart}} = J^{\top}(q)F$$

- Jacobian $\begin{bmatrix} \dot{p}_{\text{EE}} \\ \omega_{\text{EE}} \end{bmatrix} = J(q)\dot{q}$



1 kHz control loop · C++ / libfranka

- Impedance reference point

Tool centre point (TCP): controlled reference point on the EE

p_c Virtual centre of compliance (CoC) position

$r_c = p_c - p_{\text{TCP}}$ Displacement from TCP to CoC

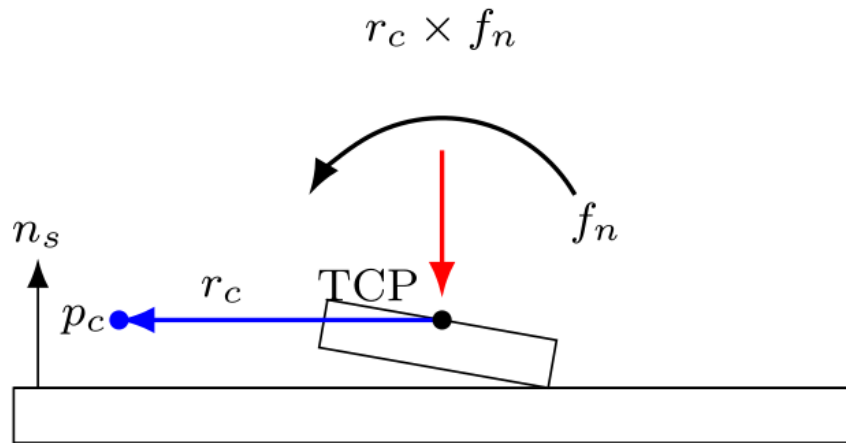
$A = \text{Ad}(r_c)$ Impedance reference: TCP $\rightarrow$ CoC

$$F_{\text{TCP}} = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{A^\top \begin{bmatrix} K_p & \mathbf{0} \\ \mathbf{0} & K_R \end{bmatrix} A}_{K_{\text{TCP}}} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{A^\top \begin{bmatrix} D_p & \mathbf{0} \\ \mathbf{0} & D_R \end{bmatrix} A}_{D_{\text{TCP}}} \begin{bmatrix} \dot{p}_d - \dot{p}_{\text{EE}} \\ \omega_d - \omega_{\text{EE}} \end{bmatrix}$$

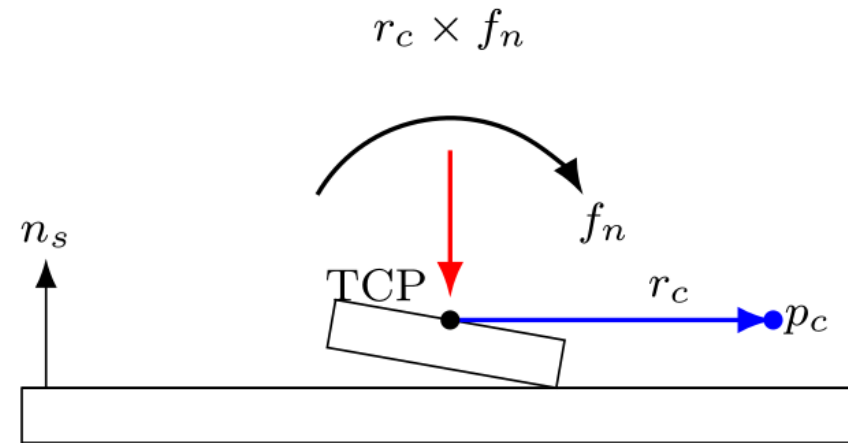
- Non-zero off-diagonal blocks in K and D couple translation and rotation.
- CoC displacement links the commanded force f and moment m .

Centre of compliance (CoC)

- Effect of CoC displacement
- Added moment can support or oppose rotation.
- Commanded force and moment can change.



(a) Supporting displacement

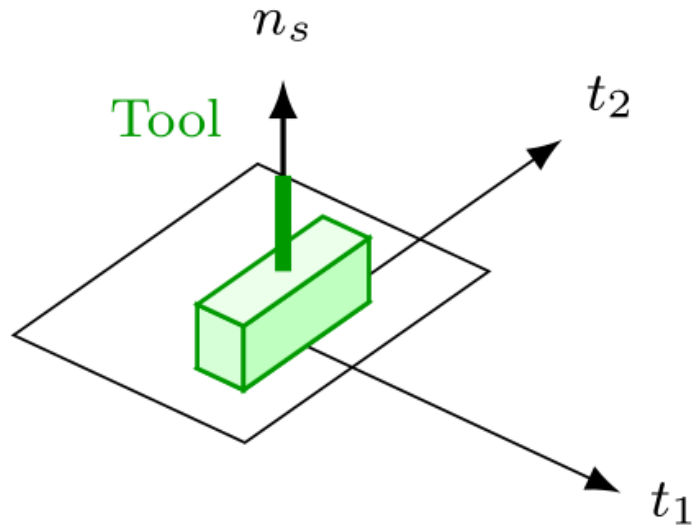


(b) Reversed displacement

- The commanded force acts as if applied at the virtual point p_c .
- Opposite displacement reverses the added moment.

Tool orientation → Surface approach → Contact establishment → Surface grinding

- Surface-relative compliance



n_s outward surface normal
 t_1, t_2 perpendicular surface tangents

- Compliant contact settings

- Normal translation

$$K_{p,n} = 350 \text{ N/m}$$

- Tangential translation

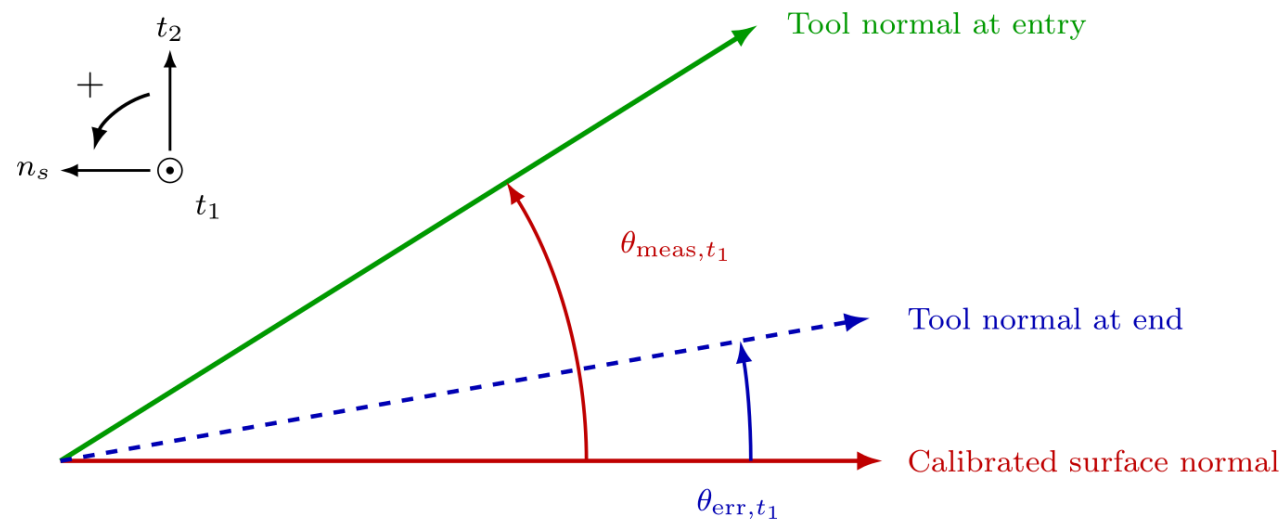
$$K_{p,t_1} = K_{p,t_2} = 2000 \text{ N/m}$$

- Rotation about the surface normal

$$K_{R,n} = 50 \text{ N m/rad}$$

- Rotations about the surface tangents

$$K_{R,t_1} = K_{R,t_2} = 5 \text{ N m/rad}$$



θ_{meas,t_1} Measured angular offset
Entry tilt relative to the
calibrated surface

θ_{err,t_1} Angular error
Tool-normal tilt relative to
the calibrated surface

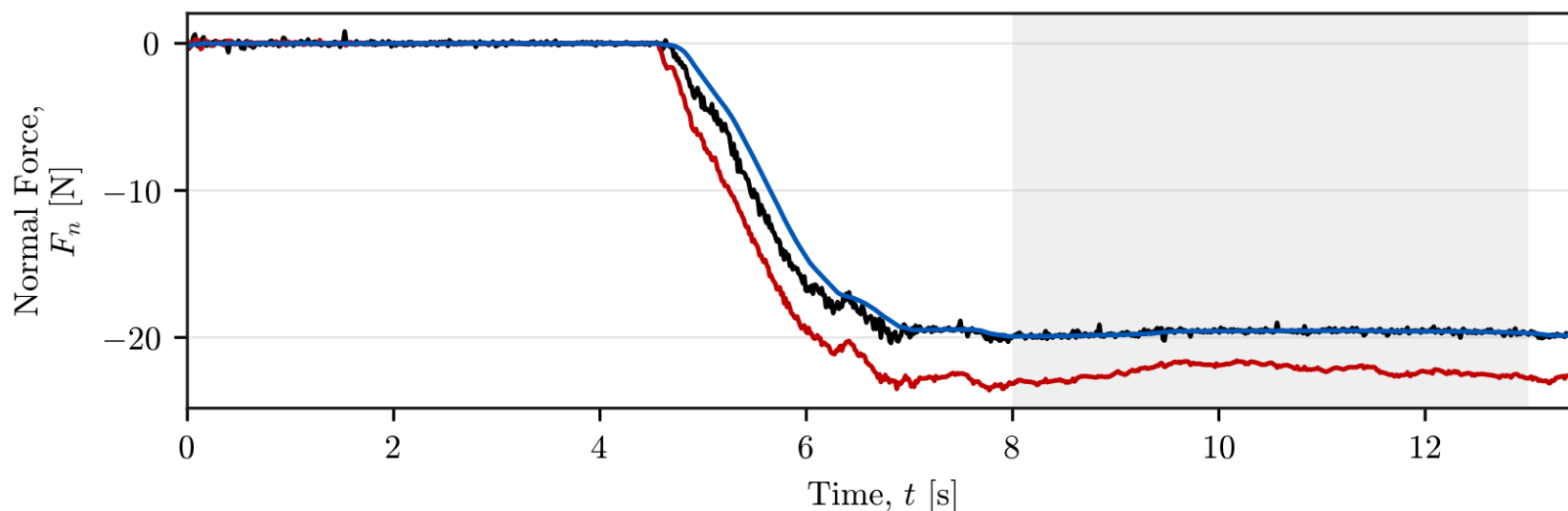
- Smaller magnitude means less angular error about the first tangent
- Positive and negative values indicate opposite tilt directions
- Calculated from measured EE orientation and the calibrated tool normal

Normal-force plausibility assessment

- Manual push: about 20 mm

Measured error change: -19.65 mm

$$\begin{aligned} F_{n,qs} &= K_{p,n} n_s^\top (e_p - e_{p,0}) \\ &= (1000 \text{ N/m}) \cdot (-19.65 \cdot 10^{-3} \text{ m}) = -19.65 \text{ N} \end{aligned}$$



— Commanded

— Model-estimated

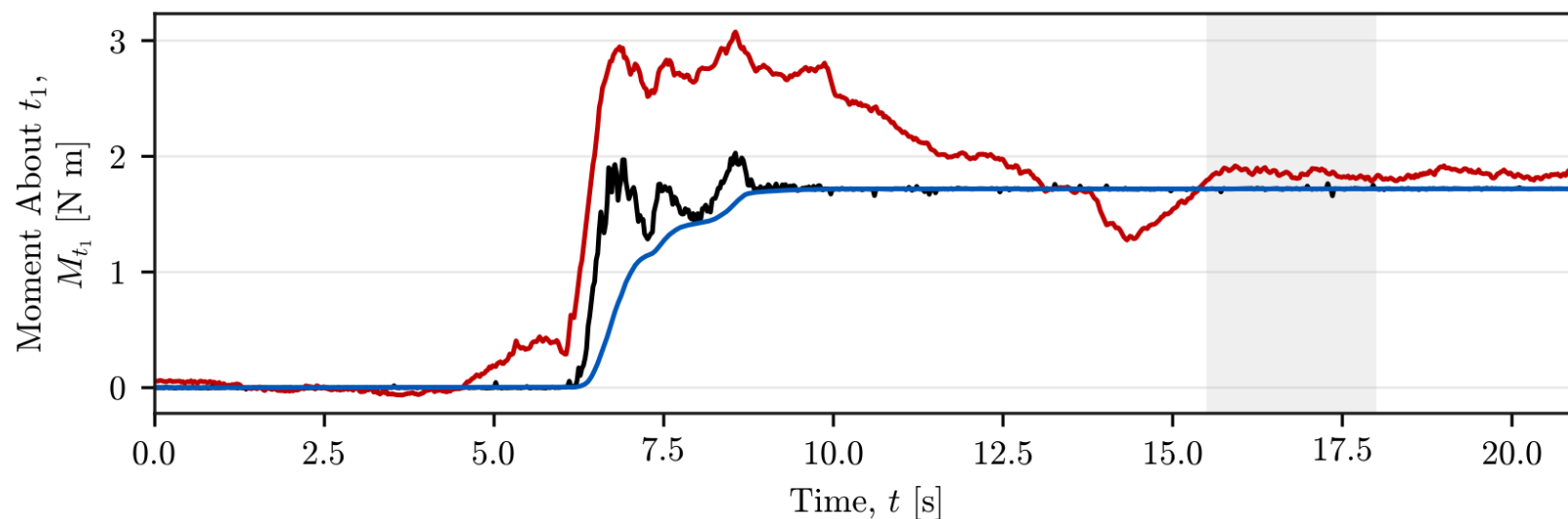
— Quasi-static prediction

- Commanded mean: -19.64 N (0.05% from calculation)
- Model-estimated mean: -22.29 N (13.50% from command)

- Manual rotation: about 6.5°

Measured error change: 6.57°

$$\begin{aligned} M_{t_1,qs} &= K_{R,t_1} t_1^\top (e_R - e_{R,0}) \\ &= (15 \text{ N m/rad}) \cdot \left(6.57 \cdot \frac{\pi}{180} \text{ rad} \right) \approx 1.72 \text{ N m} \end{aligned}$$



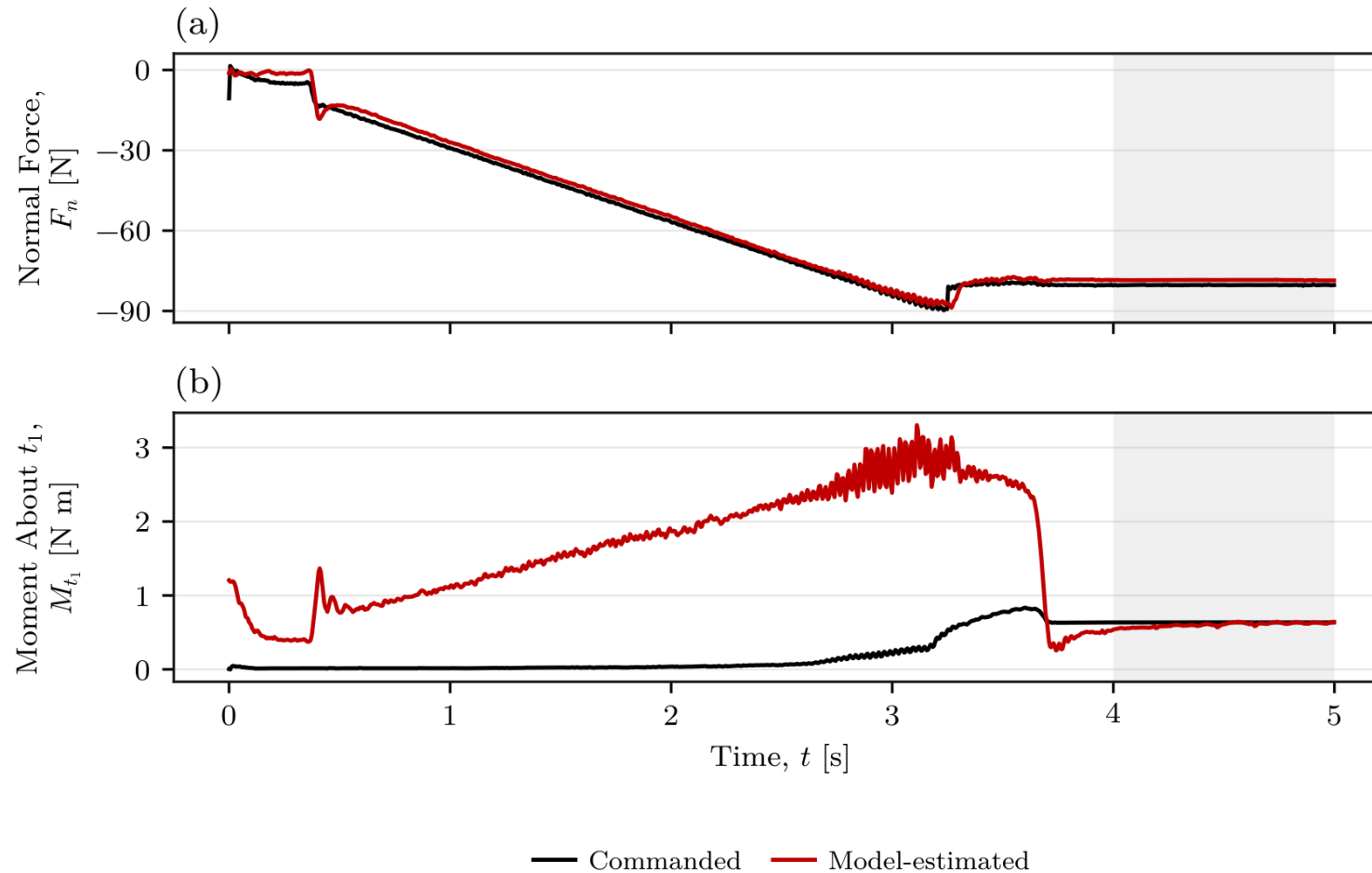
— Commanded

— Model-estimated

— Quasi-static prediction

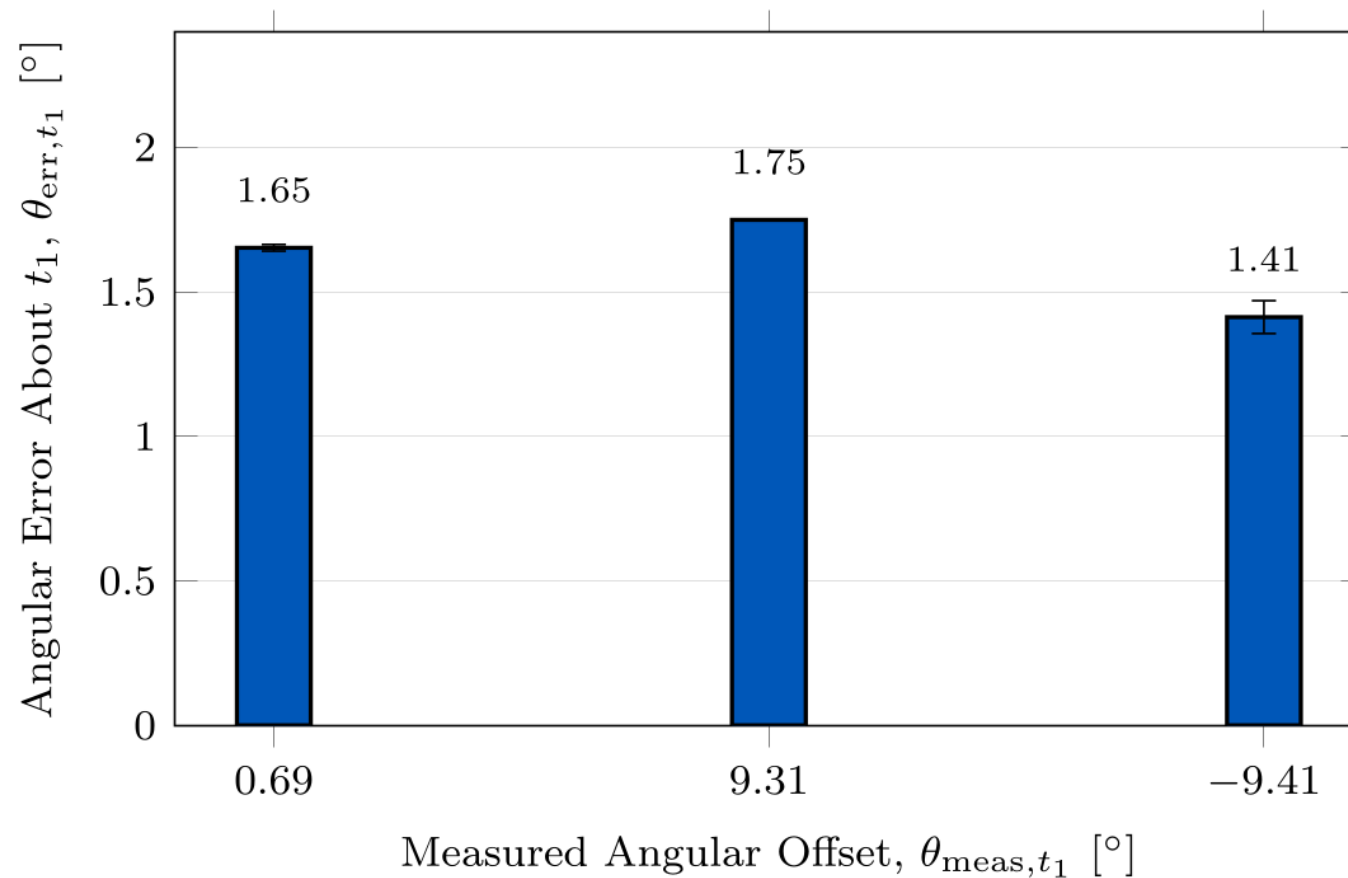
- Commanded mean: 1.72 N m (matches calculation)
- Model-estimated mean: 1.85 N m (7.75% from command)

Commanded and estimated wrench

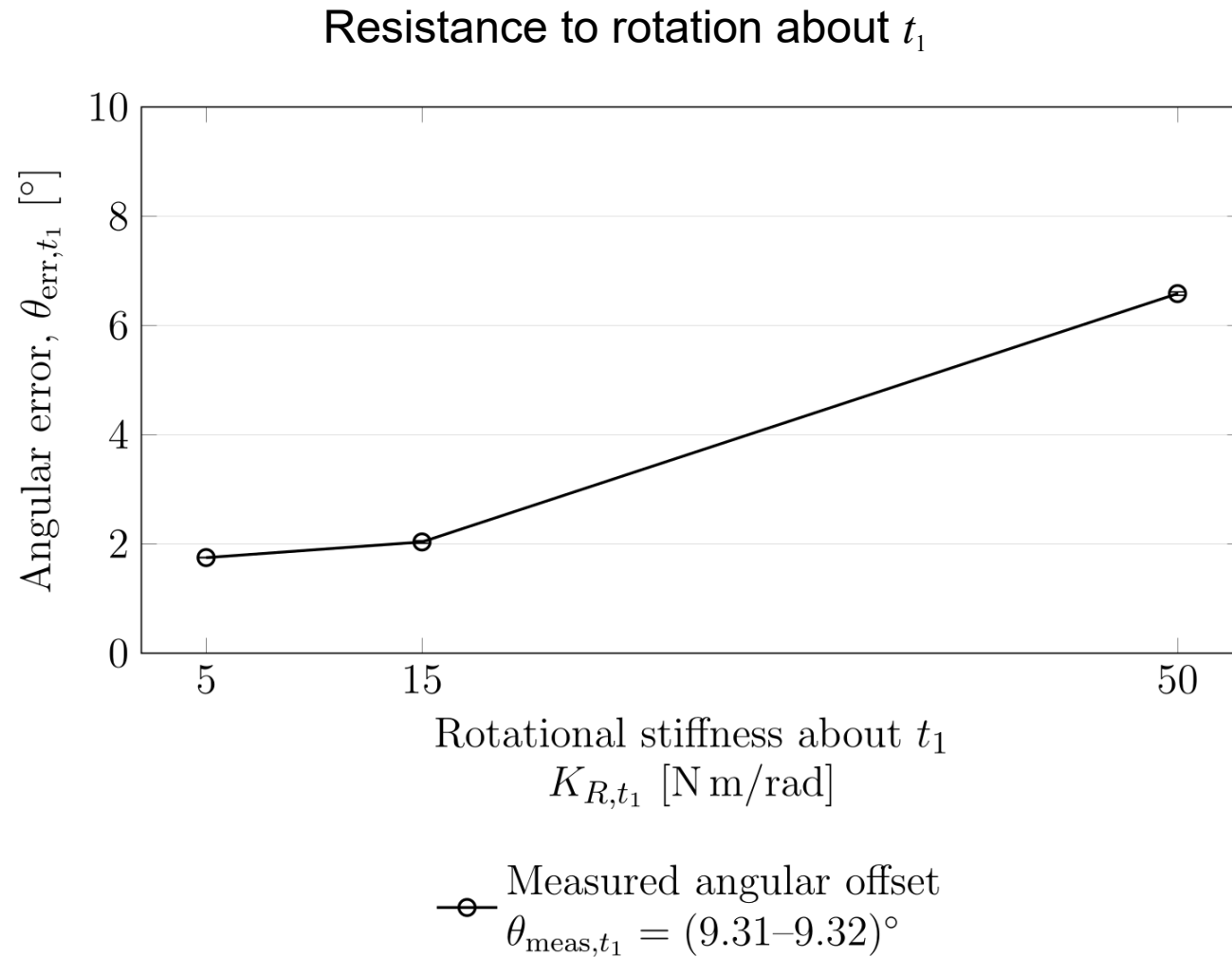


- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%

Baseline angular error



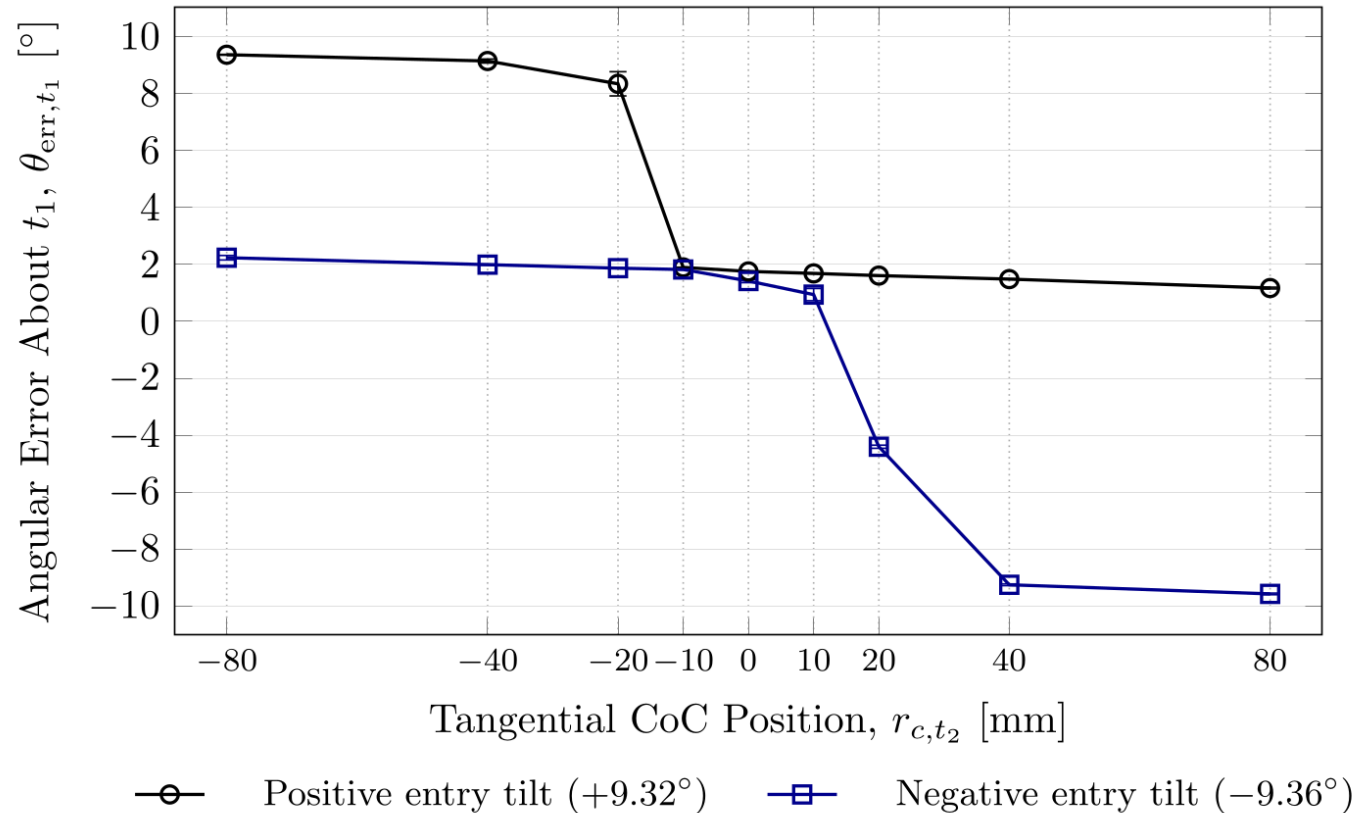
- Centre of compliance at the TCP
- Both larger entry tilts ended closer to zero
- Negative entry error changed sign



- Angular error increased from 1.75° to 6.58°

Effect of CoC position

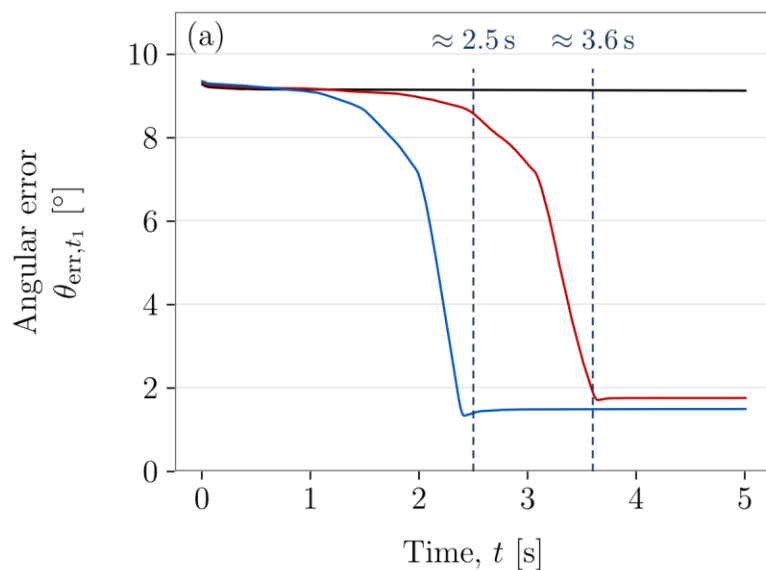
r_{c,t_2} CoC displacement along t_2



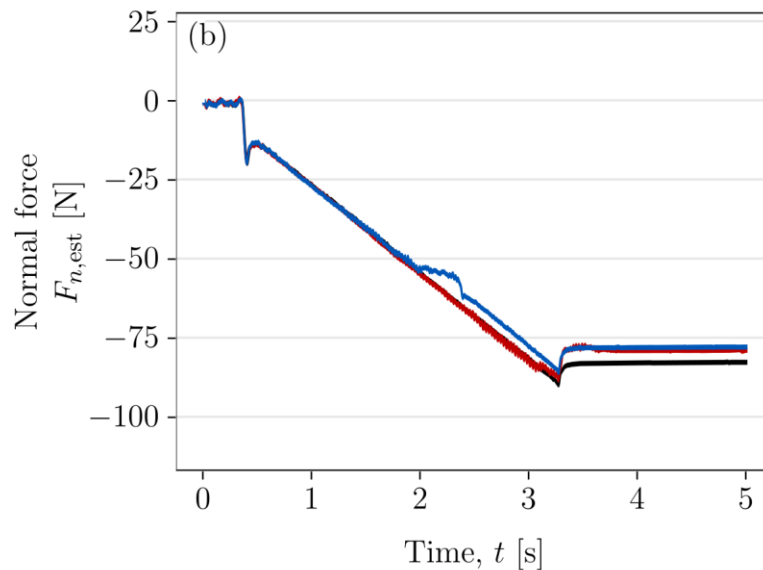
Smallest tested error magnitude

- Positive entry tilt: 1.17° at +80 mm
- Negative entry tilt: 0.94° at +10 mm
- More rotation can increase error

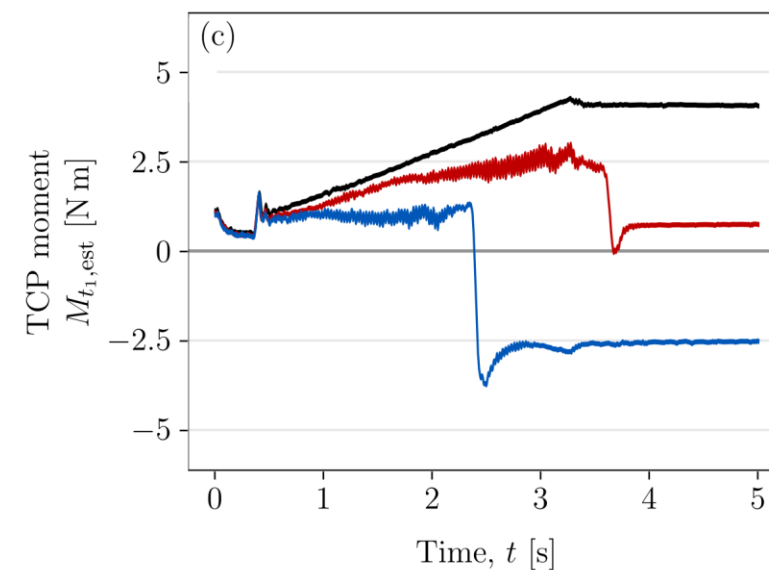
- Angular error



- Normal force



- TCP moment



— $r_{c,t_2} = -40$ mm

— TCP: $r_{c,t_2} = 0$

— $r_{c,t_2} = +40$ mm

- Reversing CoC displacement reverses the interaction moment
- Within 0.1° of angular error: 2.5 s (+40 mm), 3.6 s (TCP)
- Final normal force (approx.): -83 N (-40 mm), -79 N (TCP), -78 N (+40 mm)

- Redundancy

7 joints for a 6-DOF Cartesian pose.
One redundant direction at full rank.

$$\dim(\ker J) = 7 - 6 = 1$$

- Damping τ_d

Reduces null-space motion.
Opposes redundant joint velocity.

- Cartesian task preservation

Zero instantaneous TCP translation and rotation

$$J(q) \dot{q}_{\text{null}} = 0$$

- Conditioning τ_σ

Chooses a motion direction with a larger minimum singular value.

At a singularity, the EE loses a motion direction.

Near it, some motions need very large joint velocities.

Null-space experiment

- Commanded joint disturbance

$$\tau_{\text{dist}} = J(q)^{\top} f_{\text{dist}}$$

Joint-torque equivalent of a virtual 20 N point force on link 3.

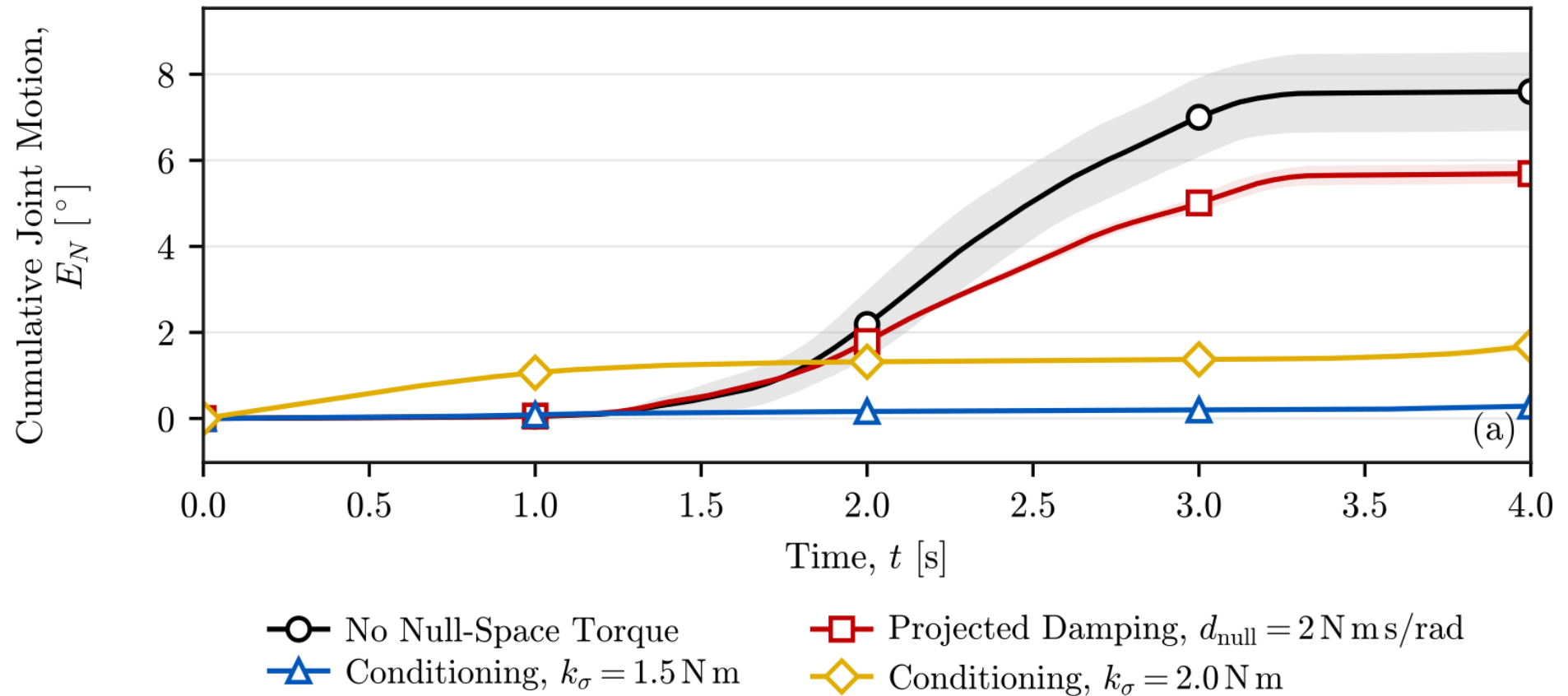
- Four settings with three trials each
 - No null-space torque
 - Damping: 2 N m s/rad

- Cartesian pose hold

Hold the captured position and orientation.

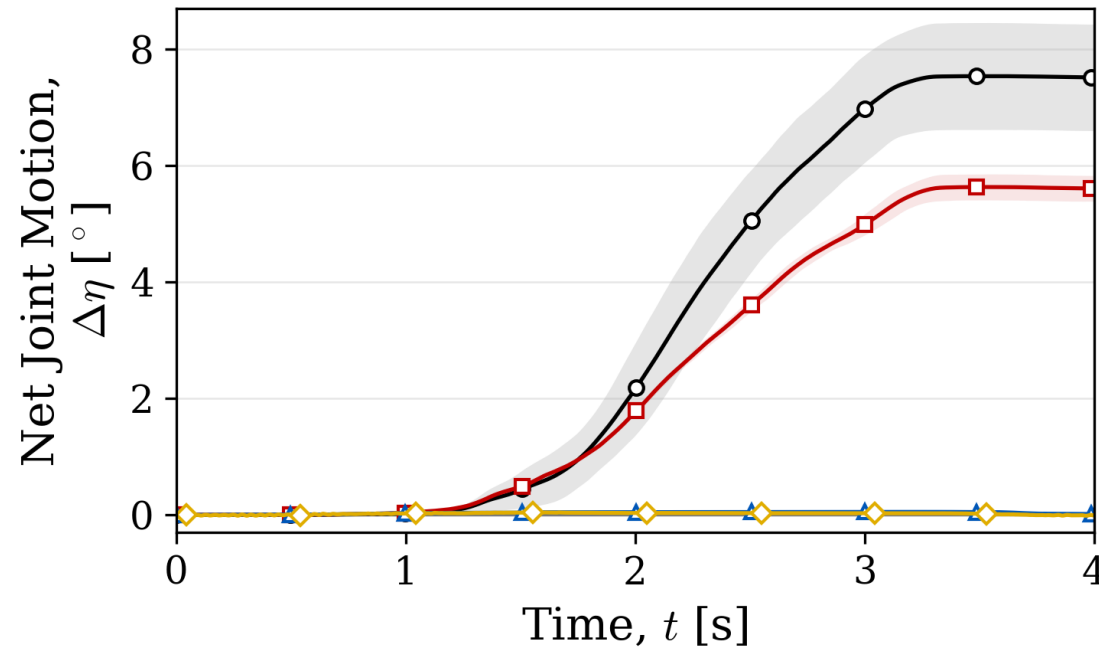
- Conditioning: 1.5 and 2.0 N m
- Cumulative and net joint motion, and conditioning

Cumulative joint motion



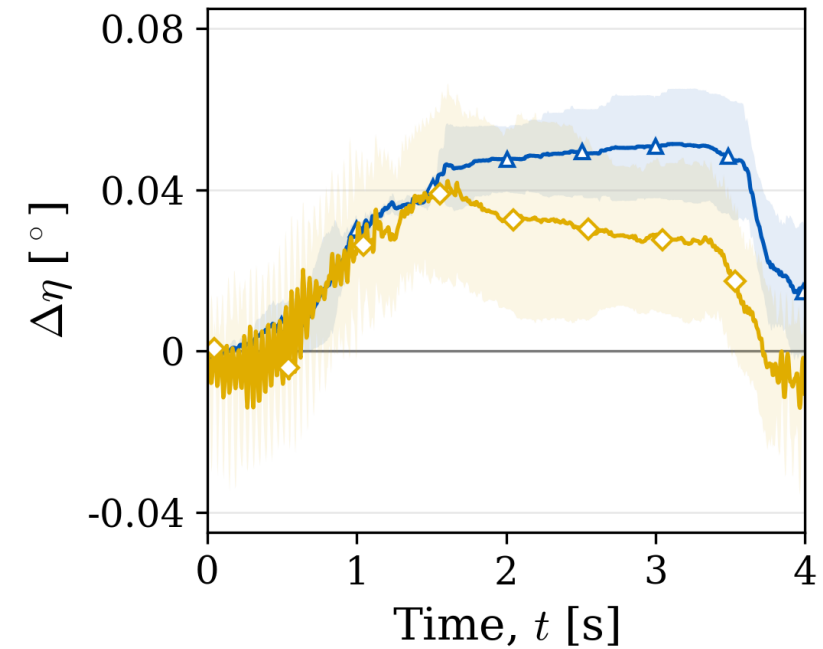
- Damping reduced cumulative joint motion from 7.60° to 5.69° (25.1%).
- With conditioning, cumulative joint motion was 0.29° and 1.69°.

- All settings



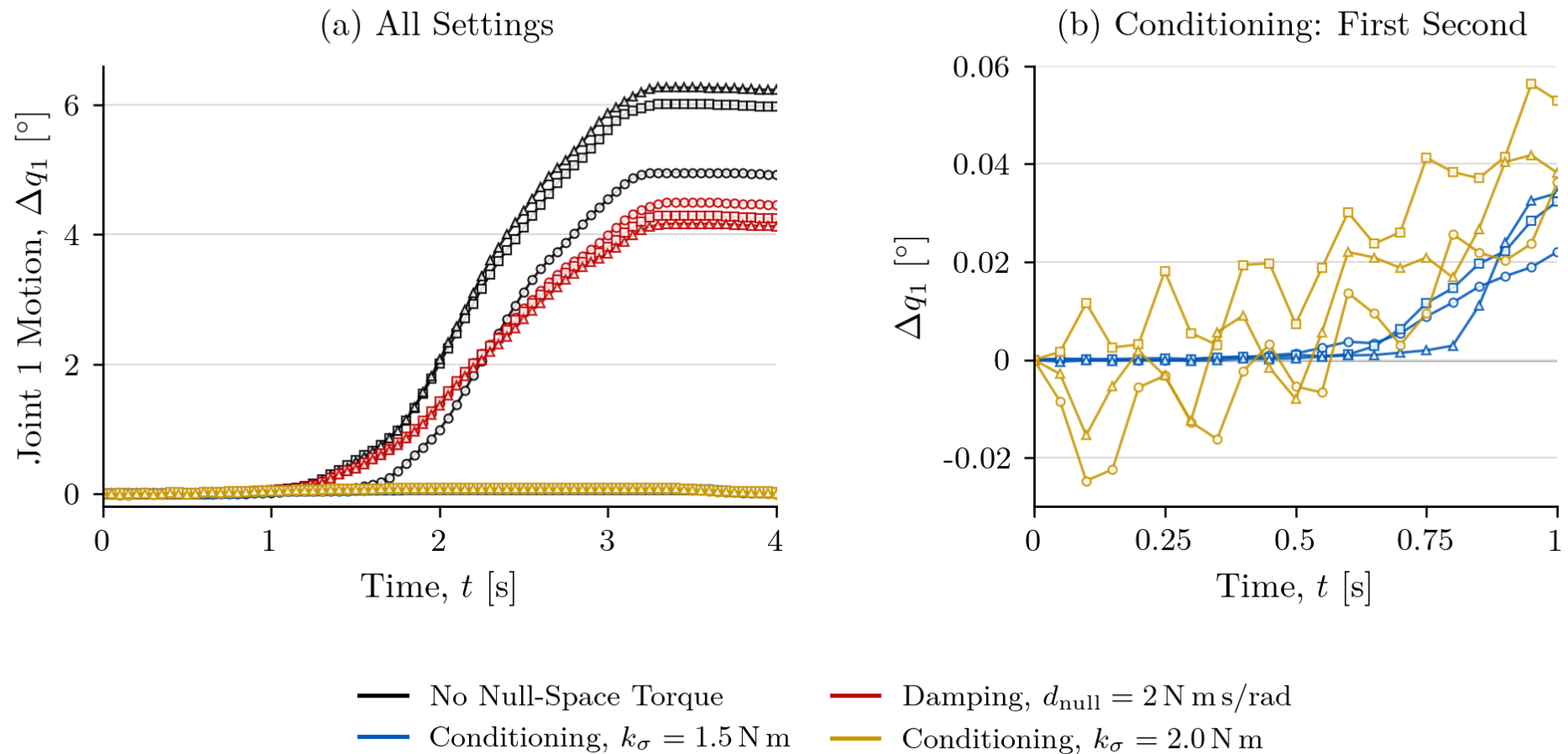
—○— No Null-Space Torque —□— Projected Damping, $d_{\text{null}} = 2 \text{ N m s/rad}$
—△— Conditioning, $k_{\sigma} = 1.5 \text{ N m}$ —◇— Conditioning, $k_{\sigma} = 2.0 \text{ N m}$

- Conditioning detail



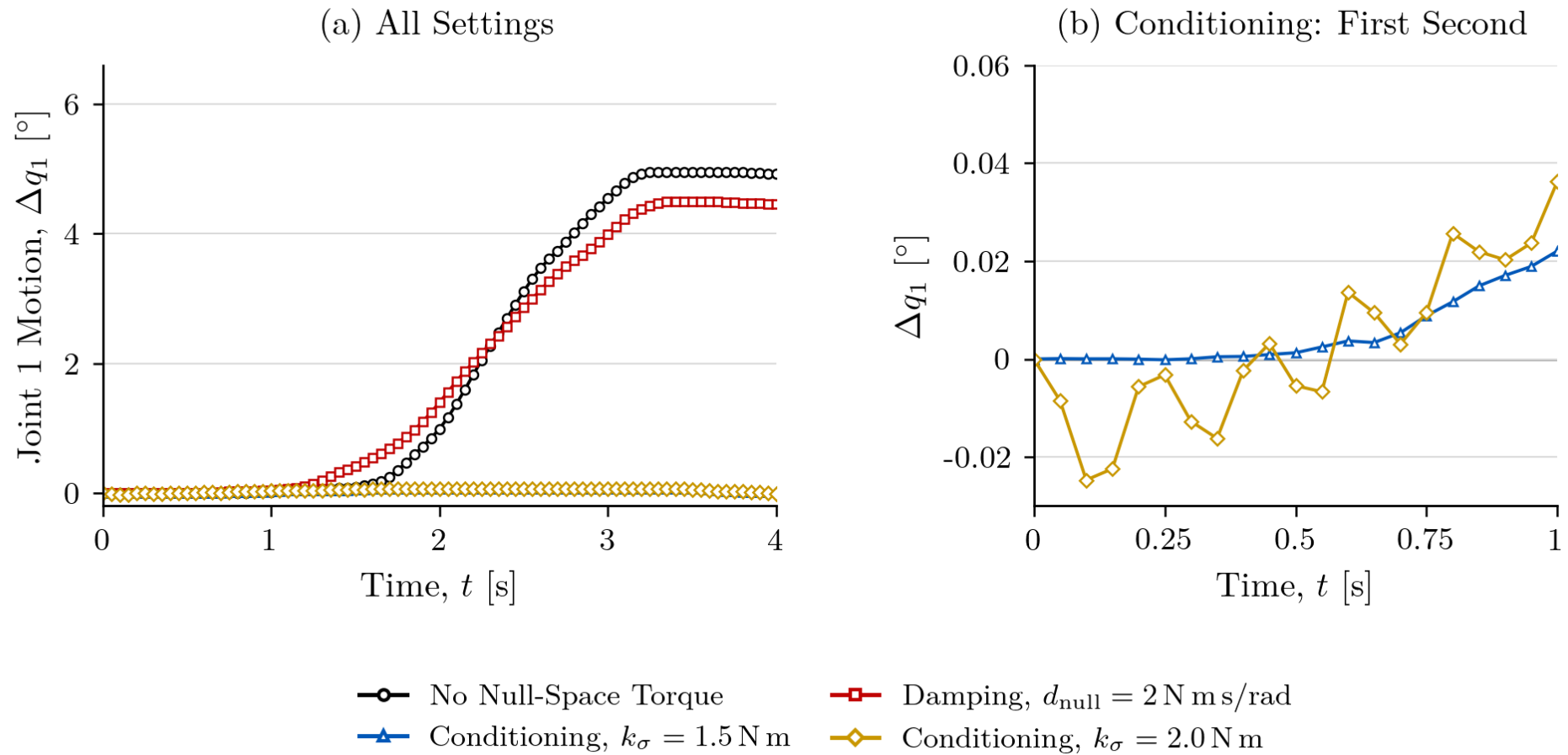
- Returning motion reduces the net joint motion.
- Conditioning ends near zero after movement in both directions.

Joint motion over time



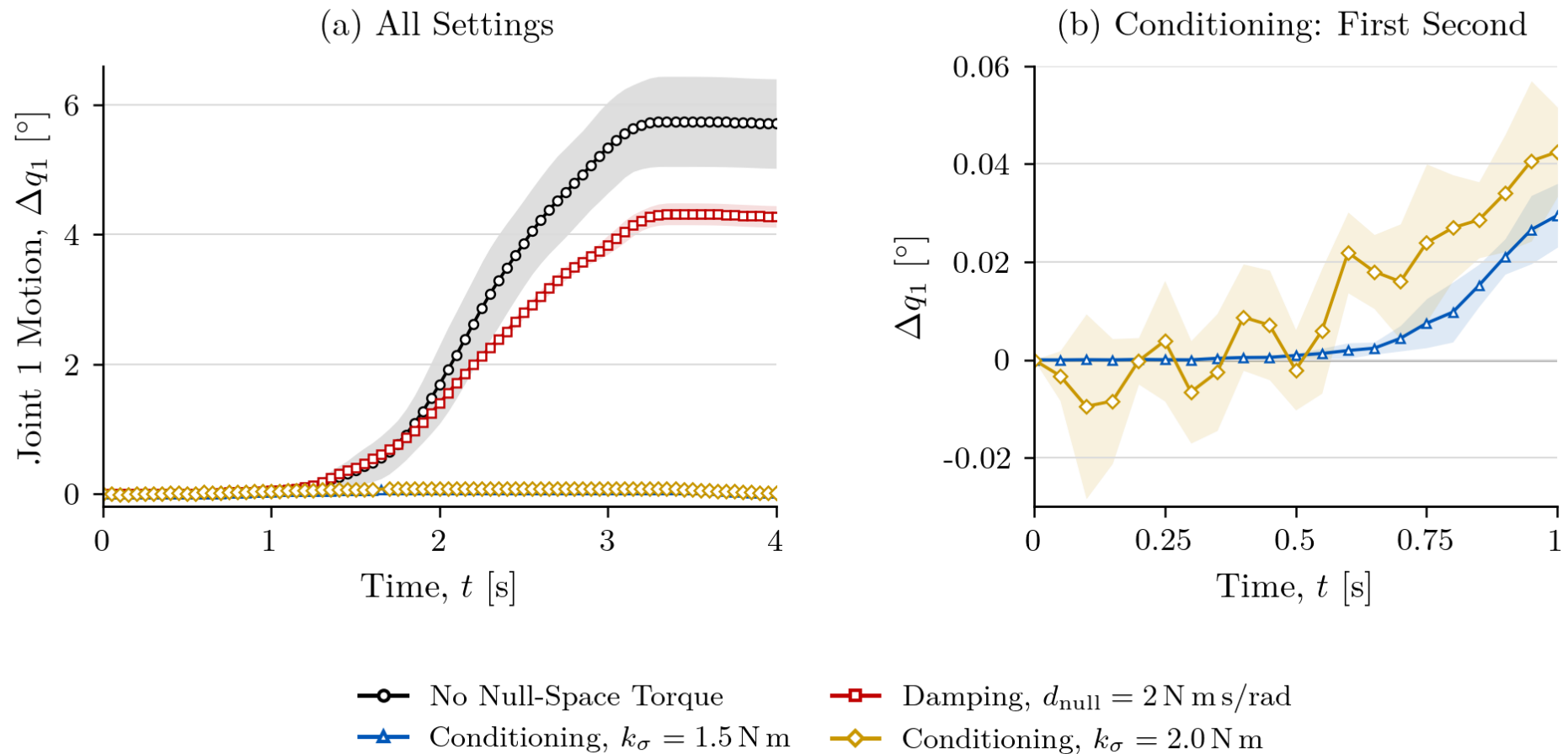
- Repeated reversals at the larger conditioning torque magnitude.
- Three individual trials per setting, each starting at zero.

Joint motion: one trial per setting

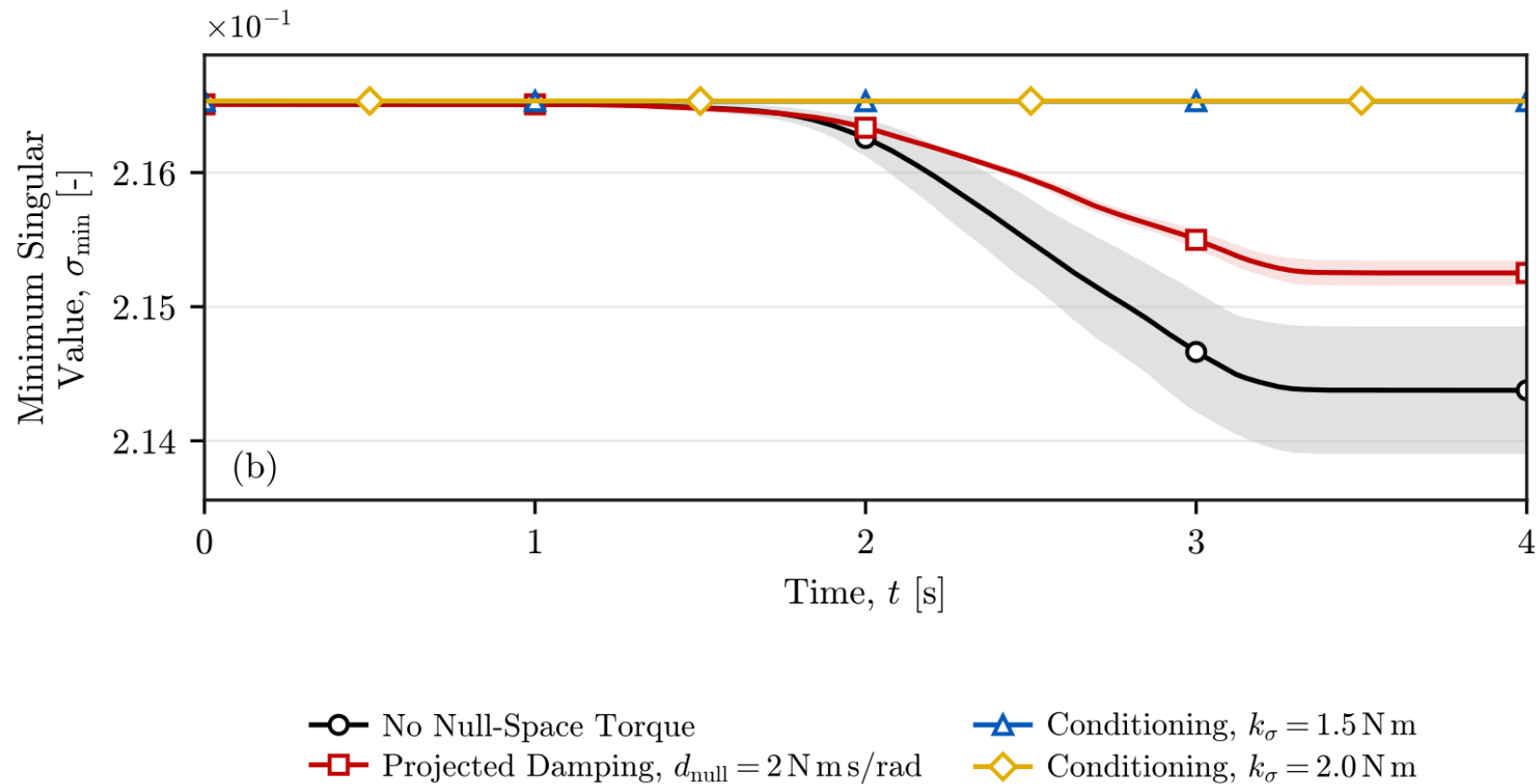


- Repetition 1 for each of the four settings.
- Repeated reversals at the larger conditioning torque magnitude.

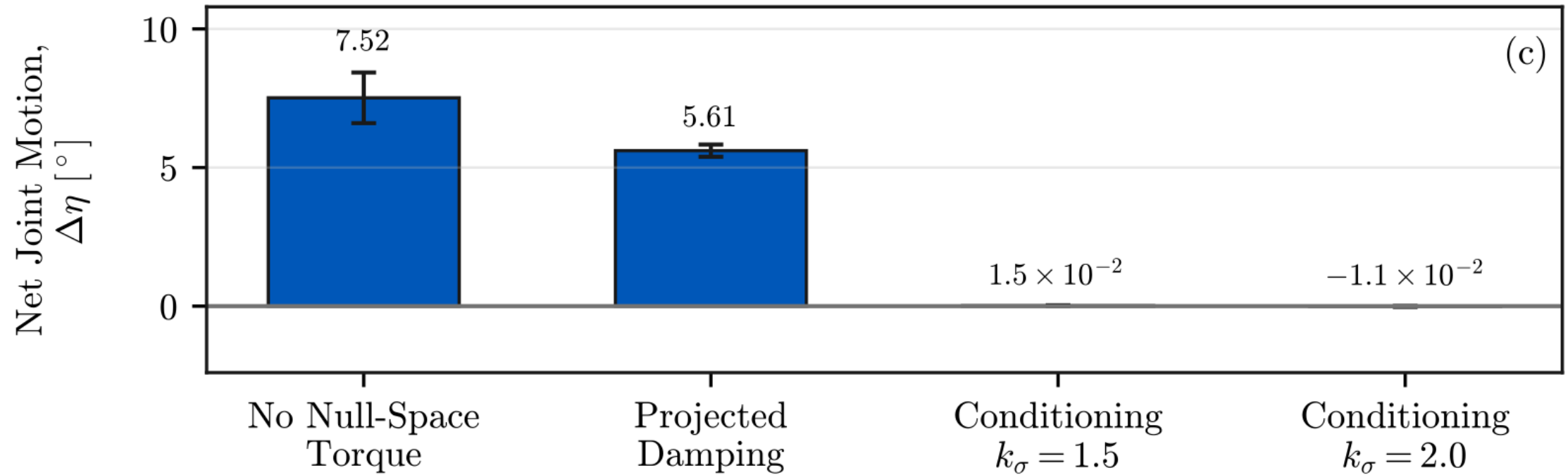
Joint motion: mean of three trials



- Three-trial mean and one sample standard deviation.
- Repeated reversals remain visible at the larger conditioning torque.



- Minimum singular value decreased with no torque and with damping.
- Both conditioning settings kept the indicator nearly constant.



- Conditioning left nearly zero net joint motion in this test.
- Small net joint motion can coexist with back-and-forth motion.

- Contact reduced angular error for the larger entry tilts
- Higher rotational stiffness left a larger angular error
- CoC displacement changes the angular error
- Damping reduced motion and conditioning countered the tested disturbance
- Future work
 - More rigid tool mount
 - Adaptive CoC selection and return to the TCP after alignment

- Null-space projector

$$N_\tau = I_7 - J^\top J^+{}^\top, \quad \tau_{\text{null}} = \tau_d + \tau_\sigma$$

Projects secondary joint torques into the kinematic null space.

J : joint $\rightarrow$ EE velocities

J^+ : pseudoinverse

I_7 : 7×7 identity matrix

- Damping torque

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

Opposes redundant joint velocity.
Zero at zero joint velocity.

d_{null} : damping coefficient [N m s/rad]

$\dot{q}$: measured joint velocity

- Conditioning torque

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

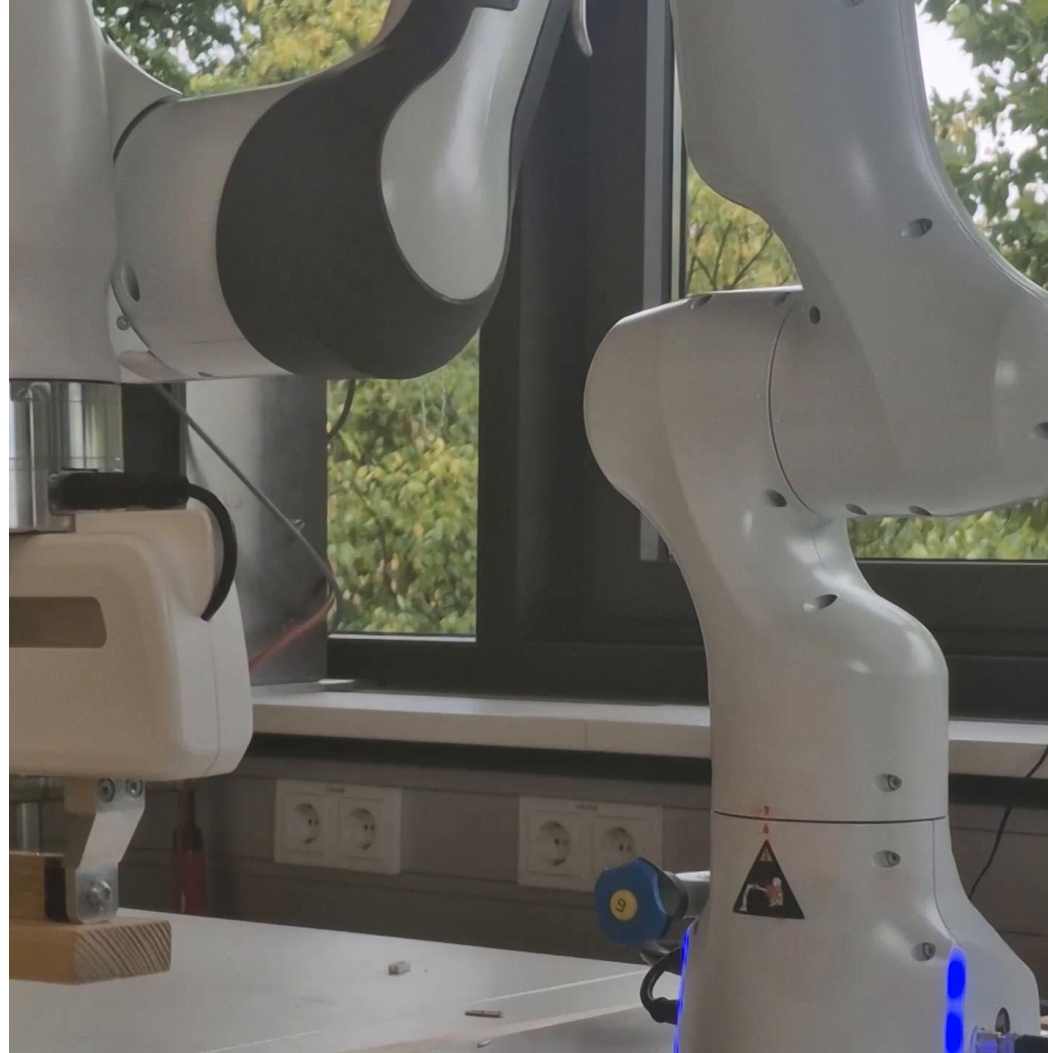
Seeks better local conditioning.
Can act at zero joint velocity.

k_σ : torque magnitude [N m]

v_7 : unit null-space direction

s_σ : direction choice, zero in deadband

Disturbance demonstration



No null-space torque → Damping → Conditioning